Math 520, F07
Exam 1

Name: Answers
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the quiz and exam.
- You must simplify your answer if possible.

Problem	Points	Score
1	15	
2	20	
3	20	
4	10	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (15 points) Given the initial-value problem

$$\frac{dy}{dt} + \frac{ty}{1+t^2} = \frac{e^{2t}}{(1+t^2)^{1/2}}, \quad y(0) = 2.$$

(a) Find an integrating factor $\mu(t)$ for this nonhomogeneous linear differential equation.

Solution: $\mu(t) = e^{\int \frac{t}{1+t^2} dt}$

$= e^{\frac{1}{2}\ln(1+t^2)}$

$= \sqrt{1+t^2}$

(b) Solve this initial-value problem based on (a).

Solution: $\frac{d}{dt}[\mu(t) y] = \mu(t)\, b(t)$

where $b(t) = \frac{e^{2t}}{(1+t^2)^{\frac{1}{2}}}$

$\int^t \frac{d}{ds}(\mu(s) y(s)) ds = \int^t \mu(s) b(s)\, ds$

$\mu(t) y(t) - \mu(0) y(0) = \int_0^t e^{2s}\, ds$

$\sqrt{1+t^2}\, y(t) - 2 = \frac{e^{2t}}{2} - \frac{1}{2}$

$y(t) = \frac{1}{\sqrt{1+t^2}}\left(\frac{e^{2t}}{2} + \frac{3}{2}\right)$

Problem 2. (20 points) Given the initial-value problem

$$\sin y \frac{dy}{dt} = \frac{-t\cos y}{1+t^2}, \qquad y(2) = \pi.$$

(a) Solve the given problem.

Solution: $-\frac{\sin y}{\cos y}\frac{dy}{dt} = \frac{t}{1+t^2}$, $\rightarrow f(y) = -\frac{\sin y}{\cos y}$, $g(t) = \frac{t}{1+t^2}$

$$\int_{y(2)}^{y(t)} -\frac{\sin r}{\cos r}\,dr = \int_2^t \frac{s}{1+s^2}\,ds$$

$$\left[\ln|\cos r|\right]_{r=\pi}^{r=y(t)} = \left[\tfrac{1}{2}\ln(1+s^2)\right]_{s=2}^{s=t}$$

$$\ln|\cos y(t)| - \ln|\cos\pi| = \tfrac{1}{2}\ln(1+t^2) - \tfrac{1}{2}\ln 5$$

$$\ln|\cos y(t)| = \ln\sqrt{\frac{1+t^2}{5}}$$

$$\cos y(t) = \pm\sqrt{\frac{1+t^2}{5}}$$

$$y(t) = \arccos\left(\pm\sqrt{\frac{1+t^2}{5}}\right)$$

Since $y(2) = \pi = \arccos(-1)$, we know that

$$y(t) = \arccos\left(-\sqrt{\frac{1+t^2}{5}}\right)$$

(b) Determine the interval of existence of the solution found in (a).

Solution: It is obvious that $\frac{1+t^2}{5} > 0$ for any t.

Furthermore, it is required that

$$\left|-\sqrt{\frac{1+t^2}{5}}\right| \le 1$$

$$\frac{1+t^2}{5} \le 1$$

$$t^2 \le 4$$

$$\Rightarrow -2 \le t \le 2$$

Problem 3. (20 points) Given a differential equation

$$e^{2at+y} + 3t^2y^2 + t + (2yt^3 + e^{2at+y})\frac{dy}{dt} = 0.$$

(a) Determine the constant a such that the equation is exact.

Solution: $M(t,y) = e^{2at+y} + 3t^2y^2 + t$, $N(t,y) = 2yt^3 + e^{2at+y}$

$$\frac{\partial M}{\partial y} = e^{2at+y} + 6t^2y, \quad \frac{\partial N}{\partial t} = 6yt^2 + 2ae^{2at+y}.$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}, \text{ i.e., the equation is exact}$$

$$\Rightarrow \quad e^{2at+y} + \cancel{6t^2y} = \cancel{6yt^2} + 2ae^{2at+y}$$

$$(2a-1)e^{2at+y} = 0$$

$$2a - 1 = 0 \Rightarrow a = \frac{1}{2}$$

(b) Find the general solution of the resulting equation from (a).

Solution: $M(t,y) = e^{t+y} + 3t^2y^2 + t = \frac{\partial\phi}{\partial t}$

$$\Rightarrow \phi(t,y) = \int M(t,y)\,dt + h(y)$$

$$= \int (e^{t+y} + 3t^2y^2 + t)\,dt + h(y)$$

$$= e^{t+y} + t^3y^2 + \frac{t^2}{2} + h(y)$$

$$N(t,y) = 2yt^3 + e^{t+y} = \frac{\partial\phi}{\partial y}$$

$$\Rightarrow \phi(t,y) = \int N(t,y)\,dy + k(t)$$

$$= \int (2yt^3 + e^{t+y})\,dy + k(t)$$

$$= y^2t^3 + e^{t+y} + k(t)$$

$$\Rightarrow h(y) = 0, \; k(t) = \frac{t^2}{2} \Rightarrow \phi(t,y) = yt^3 + e^{t+y} + \frac{t^2}{2}$$

Thus the general solution of this equation is given by

$$\underline{yt^3 + e^{t+y} + \frac{t^2}{2} = C}$$

Problem 4. (10 points) Given the differential equation

$$4ty + \frac{3}{2}y^2 + (t^2 + ty)\frac{dy}{dt} = 0.$$

Find an integrating factor $\mu(t)$ such that $\mu(t)(4ty + \frac{3}{2}y^2) + \mu(t)(t^2 + ty)\frac{dy}{dt} = 0$ is exact.

Solution: $M(t,y) = 4ty + \frac{3}{2}y^2$

$N(t,y) = t^2 + ty.$

$$\frac{\partial M}{\partial y} = 4t + 3y$$

$$\frac{\partial N}{\partial t} = 2t + y$$

Thus, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial t}$, i.e., this equation is not exact.

Fortunately, we have

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial t}}{N} = \frac{2t+2y}{t^2+ty} = \frac{2}{t} = R(t)$$

is a function of t alone. So that

$$\mu(t) = e^{\int R(t)\,dt} = e^{\int \frac{2}{t}dt}$$

$$= e^{\ln t^2} = t^2$$

is an integrating factor for this equation.

Problem 5. (20 points) Given the initial-value problem

$$y' = e^{-t} + y^2, \quad y(0) = -1.$$

(a) Compute the first two Picard iterates for the above problem.

Solution: $f(t,y) = e^{-t} + y^2$

$$y_n(t) = y(t_0) + \int_{t_0}^{t} f(s, y_{n-1}(s))\,ds$$
$$= -1 + \int_0^t (e^{-s} + y_{n-1}^2(s))\,ds$$

$$y_1(t) = -1 + \int_0^t (e^{-s} + (-1)^2)\,ds$$
$$= -1 + (-e^{-t} + 1) + t = -e^{-t} + t$$

$$y_2(t) = -1 + \int_0^t (e^{-s} + y_1^2(s))\,ds$$
$$= -1 + \int_0^t (e^{-s} + (-e^{-s} + s)^2)\,ds$$
$$= -1 + \int_0^t (e^{-s} + e^{-2s} - 2se^{-s} + s^2)\,ds$$
$$= -1 + (-e^{-t} + 1) + \left(-\frac{e^{-2t}}{2} + \frac{1}{2}\right) + \left(2e^{-t}(t+1) - 2\right) + \frac{t^3}{3}$$
$$= 2te^{-t} - \frac{1}{2}e^{-2t} + e^{-t} + \frac{1}{3}t^3 - \frac{3}{2}$$

(b) Show that the solution of the given problem exists on the interval $0 \le t \le \frac{1}{2+2\sqrt{2}}$.

Solution: Here $t_0 = 0$, $y_0 = -1$, $a = \frac{1}{2+2\sqrt{2}}$. Let R be the rectangle $0 \le t \le \frac{1}{2+2\sqrt{2}}$, $|y+1| \le b$. Then

$$M = \max_{(t,y)\in R} |f(t,y)| = \max_{\substack{0 \le t \le \frac{1}{2+2\sqrt{2}} \\ |y+1| \le b}} |e^{-t} + y^2|$$

$$= 1 + (-1-b)^2$$
$$= 2 + 2b + b^2$$

$$\alpha = \min\left(a, \frac{b}{M}\right) = \min\left(\frac{1}{2+2\sqrt{2}}, \frac{b}{2+2b+b^2}\right)$$
$$= \min\left(\frac{1}{2+2\sqrt{2}}, \frac{1}{2+\frac{2}{b}+b}\right)$$

$\frac{2}{b} + b$ is minimized when $\frac{2}{b} = b$, i.e. $b^2 = 2$, $b = \sqrt{2}$. Let us choose $b = \sqrt{2}$, then we get

$$\alpha = \min\left(\frac{1}{2+2\sqrt{2}}, \frac{1}{2+\sqrt{2}+\sqrt{2}}\right) = \frac{1}{2+2\sqrt{2}}$$

Thus, $y(t)$ exists on $0 \le t \le \frac{1}{2+2\sqrt{2}}$ by Theorem 2.

Problem 6. (15 points) Given the second-order linear differential equation $y'' + y = 0$.
(a) Show that $y_1 = \sin t$ and $y_2 = \cos t$ are solutions of the above equation.

Solution: $L[y](t) = y'' + y$.

$$L[\sin t](t) = (\sin t)'' + \sin t$$
$$= -\sin t + \sin t = 0$$
$$L[\cos t](t) = (\cos t)'' + \cos t$$
$$= -\cos t + \cos t = 0$$

So that $y_1(t) = \sin t$ and $y_2(t) = \cos t$ are the solutions.

(b) Compute $W[y_1, y_2](t)$ and show that $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of the given equation on the interval $[-\infty, \infty]$.

Solution: $W[y_1, y_2](t) = y_1(t) y_2'(t) - y_1'(t) y_2(t)$

$$= \sin t(-\sin t) - \cos t \cdot \cos t$$
$$= -(\sin^2 t + \cos^2 t) = -1 \neq 0$$

So that $c_1 \sin t + c_2 \cos t$ is the general solution of the above equation.

(c) Solve the initial-value problem $y'' + y = 0, y(0) = -1, y'(0) = 2$.

Solution: $y(t) = c_1 \sin t + c_2 \cos t$, we need to determine c_1 and c_2 by the initial condition.

$$y(0) = c_1 \sin(0) + c_2 \cos(0) = c_2 = -1$$
$$y'(0) = c_1 \cos(0) - c_2 \sin(0) = c_1 = 2$$

$$\Rightarrow c_2 = -1, \ c_1 = 2$$

Thus $y(t) = 2\sin t - \cos t$